Problem Set 2

Probability Theory for Computer Science - Week 2

☺ Basic ! Important ★ Challenging (with hints)

☺ *Basic level*

☺ 1. *API response model.* An AI service request ends in success, timeout, or invalid output with probabilities 0.72, 0.18, and 0.10.

(a) Check the non-negativity and normalization axioms.

(b) Find the probability that the request does not succeed.

(c) Let R be the event that the client retries after timeout or invalid output. Find $P(R)$.

☺ 2. *Complement of model availability.* Let A be the event that a deployed model is available during a randomly selected minute. Suppose $P(A) = 0.997$.

(a) State the complement A^c in words.

(b) Use the complement rule to find $P(A^c)$.

(c) Explain why $P(A) + P(A^c) = 1$ follows from the axioms.

☺ 3. *Disjoint compiler outcomes.* A generated program either compiles on the first attempt C, fails with a syntax error S, or fails with a type error T. These events are pairwise disjoint and exhaustive. Suppose $P(C) = 0.61$ and $P(S) = 0.24$.

(a) Find $P(T)$.

(b) Find $P(S \cup T)$.

(c) Why may the probabilities of S and T be added directly?

☺ 4. *Nested safety events.* Let H be the event that a generated answer contains harmful content, and let F be the event that it contains harmful content that passes the safety filter. Thus $F \subseteq H$. Suppose $P(H) = 0.03$ and $P(F) = 0.004$.

(a) Check the monotonicity inequality for F and H.

(b) Find $P(H \setminus F)$.

(c) Interpret $H \setminus F$ in the context of the filter.

☺ 5. *Dataset audit categories.* Every record in an audit belongs to exactly one category: clean C, duplicate D, or mislabeled M. The probabilities are $P(C) = 0.84$, $P(D) = 0.09$, and $P(M) = 0.07$.

(a) Verify that these atom probabilities define a probability model.

(b) Find the probability that a record needs correction.

(c) Find $P((D \cup M)^c)$.

☺ 6. *Cache result.* A request is served from exactly one of three sources: memory cache M, disk cache D, or origin server O. The corresponding probabilities are 0.45, 0.35, and 0.20.

(a) Find $P(M \cup D)$.

(b) Find $P((M \cup D)^c)$.

(c) Which probability axiom justifies adding $P(M)$ and $P(D)$?

☺ 7. *Confidence threshold events.* For a classifier score X, define $A = \{X \geq 0.50\}$, $B = \{X \geq 0.70\}$, and $C = \{X \geq 0.90\}$. Suppose $P(A) = 0.80$, $P(B) = 0.55$, and $P(C) = 0.18$.

(a) Write the inclusion relationships among A, B, and C.

(b) Verify that the probabilities are consistent with monotonicity.

(c) Find $P(B \setminus C)$.

☺ 8. *Model-version traffic split.* Traffic is sent to model versions v1, v2, and v3 with probabilities 0.20, 0.50, and 0.30. Let $E = \{v1, v3\}$.

(a) Use finite additivity to find $P(E)$.

(b) Find $P(E^c)$.

(c) List all events in the power set of $\Omega = \{v1, v2, v3\}$.

☺ 9. *Spot the invalid model.* For $\Omega = \{\text{fast}, \text{medium}, \text{slow}\}$, three proposed probability vectors are (0.50, 0.30, 0.20), (0.60, 0.50, -0.10), and (0.40, 0.40, 0.40).

(a) Determine which vector defines a probability model.

(b) For each invalid vector, name the violated axiom.

(c) Can a probability assigned to an event exceed 1? Derive your answer from monotonicity and $P(\Omega) = 1$.

! *Important problems*

! 10. *Overlapping answer-quality errors.* In a language-model audit, H is the event of a hallucination and C is the event of a citation error. Suppose $P(H) = 0.11$, $P(C) = 0.08$, and $P(H \cap C) = 0.03$.

(a) Find $P(H \cup C)$.

(b) Find the probability of neither error.

(c) Find the probability of exactly one of the two errors.

(d) Find $P(H \setminus C)$ and $P(C \setminus H)$.

! 11. *Three-way benchmark failure.* A benchmark records reasoning failure R, formatting failure F, and tool-use failure T. Suppose $P(R)=0.12$, $P(F)=0.09$, $P(T)=0.07$, $P(R \cap F)=0.03$, $P(R \cap T)=0.02$, $P(F \cap T)=0.01$, and $P(R \cap F \cap T)=0.005$.

(a) Use inclusion-exclusion to find $P(R \cup F \cup T)$.

(b) Find the probability of no recorded failure.

(c) Find the probability that exactly R and F occur but T does not.

! 12. *Union bound for automated checks.* A software release can fail a security check S, a performance check P, or a compatibility check C. You know only that $P(S)=0.02$, $P(P)=0.05$, and $P(C)=0.04$.

(a) Use the union bound to give an upper bound for the probability that at least one check fails.

(b) Give a lower bound for the probability that all three checks pass.

(c) Why is 0.11 not necessarily the exact failure probability?

! 13. *Feasible intersection range.* Two monitoring systems flag anomalous traffic. Let A and B be their flag events, with $P(A)=0.40$ and $P(B)=0.55$.

(a) Use $P(A \cup B) \leq 1$ to obtain a lower bound for $P(A \cap B)$.

(b) Use monotonicity to obtain an upper bound for $P(A \cap B)$.

(c) Give the full feasible interval for $P(A \cap B)$.

(d) For each endpoint, find $P(A \cup B)$.

! 14. *Partitioning inference requests.* Inference requests are partitioned into text T, image I, and audio A requests. Suppose $P(T)=0.58$, $P(I)=0.27$, and the remaining requests are audio. A premium request event P consists of disjoint parts $P \cap T$, $P \cap I$, and $P \cap A$ with probabilities 0.12, 0.08, and 0.03.

(a) Find $P(A)$.

(b) Find the probability of a premium request using the partition.

(c) Find the probability of a non-premium request.

! 15. *Probability proportional to compute weight.* A scheduler selects one of four jobs $J_1,\dots,J_4$. The selection probability is proportional to the weights 1, 2, 3, and 4.

(a) Find the normalizing constant c when $P(J_i) = ci$.

(b) Find the probability that J_3 or J_4 is selected.

(c) Find the probability that an even-indexed job is selected.

! 16. *Symmetric difference in model drift.* Let A be the event that a sample is flagged by yesterday's model and B the event that it is flagged by today's model. Suppose $P(A)=0.18$, $P(B)=0.22$, and $P(A \cap B)=0.12$.

(a) The symmetric difference $A \triangle B$ contains samples flagged by exactly one model. Express it using set difference.

(b) Find $P(A \triangle B)$.

(c) Find the probability that the two models agree on flag status.

! 17. *Nested latency thresholds.* Let L_{100}, L_{200}, and L_{500} be the events that latency exceeds 100, 200, and 500 ms. Their probabilities are 0.25, 0.10, and 0.015.

(a) Write the inclusion relationships among the three events.

(b) Find the probability that latency is above 100 ms but at most 200 ms.

(c) Find the probability that latency is above 200 ms but at most 500 ms.

(d) Find the probability that latency is at most 100 ms.

! 18. *Countable token-length model.* A generated identifier has positive integer length N with $P(N=n)=c/2^n$, $n=1,2,\ldots$.

(a) Use countable additivity to find c.

(b) Find $P(N \le 4)$.

(c) Find $P(N > 8)$.

(d) Show that $P(N > k)$ tends to 0 as k tends to infinity.

! 19. *Bounds from partial audit data.* In a dataset, D is the event of a duplicate and M the event of a missing label. You know $P(D)=0.14$ and $P(D \cup M)=0.21$, but $P(M)$ is not reported.

(a) Express $P(M)$ in terms of $P(D \cap M)$.

(b) Use $0 \le P(D \cap M) \le P(D)$ to give the tightest possible range for $P(M)$.

(c) Interpret the two endpoint cases.

★ *Challenging problems*

★ 20. *When is the union bound exact?.* For events $A_1,\ldots,A_n$, the union bound states $P(\cup_{i=1}^{n} A_i) \le \Sigma_{i=1}^{n} P(A_i)$.

(a) Show that equality holds when the events are pairwise disjoint.

(b) For n=2, prove that equality implies $P(A_1 \cap A_2)=0$.

(c) Give a computing example in which two events overlap as sets but equality still holds.

Hint. For the last part, let the overlap be a possible event of probability zero. Set equality is stronger than equality up to a null event.

★ 21. *Frechet bounds for two detectors.* Two AI detectors raise flags A and B with probabilities a and b. No information about their dependence is available.

(a) Prove $\max(0,a+b-1) \le P(A \cap B) \le \min(a,b)$.

(b) Use inclusion-exclusion to derive bounds for $P(A \cup B)$.

(c) For a=0.65 and b=0.45, compute both intervals.

(d) Construct finite probability models that attain each endpoint of the intersection interval.

Hint. Partition Ω into $A \cap B$, $A \setminus B$, $B \setminus A$, and neither. Write each atom probability in terms of $x=P(A \cap B)$, then require every atom to be nonnegative.

★ 22. *Continuity from below for expanding logs.* Let A_n be the event that a security incident appears among the first n log entries. Then $A_1 \subseteq A_2 \subseteq \ldots$ and $A=\cup_{n=1}^{\infty} A_n$.

(a) Define disjoint events $B_1=A_1$ and $B_n=A_n \setminus A_{n-1}$ for $n \geq 2$.

(b) Show that $A_n=\cup_{k=1}^{n} B_k$ and $A=\cup_{k=1}^{\infty} B_k$.

(c) Use countable additivity to prove $P(A_n)$ tends to $P(A)$.

Hint. The B_n are the newly added outcomes. Convert both probabilities into partial and infinite sums of $P(B_n)$.

★ 23. *Telescoping bug-rank model.* The rank N of the first relevant bug report has probability $P(N=n)=c/[n(n+1)]$ for $n=1,2,\ldots.$

(a) Use $1/[n(n+1)]=1/n-1/(n+1)$ to find c.

(b) Find $P(N > k)$.

(c) Find P(N is even). Express the answer using an alternating harmonic series.

(d) Check directly that $P(N \leq k)+P(N > k)=1$.

Hint. Telescope the finite partial sums before taking a limit. For the even case, sum $1/(2m)-1/(2m+1)$ and relate it to ln 2.

★ 24. *Feasibility of three audit flags.* Events H, C, and S denote hallucination, citation error, and safety violation. Suppose each has probability 0.20, every pairwise intersection has probability 0.09, and the triple intersection has probability t.

(a) Use inclusion-exclusion to express $P(H \cup C \cup S)$ in terms of t.

(b) Write the probability of exactly H and C, but not S, in terms of t.

(c) Use non-negativity of all eight atoms to determine the feasible interval for t.

(d) Find the corresponding range of $P(H \cup C \cup S)$.

Hint. Draw the eight disjoint regions of a three-set Venn diagram. The strongest lower bound comes from the 'none' atom or a one-event-only atom; check all of them.

★ 25. *Infinite seed model.* A randomized test chooses a positive integer seed K with $P(K=k)=2^{-k}$. Let A be the event that K is a power of 2 and B the event that K is even.

(a) Verify normalization using countable additivity.

(b) Find P(B).

(c) Write P(A) as an explicit infinite series and bound it between two simple rational numbers.

(d) Find $P(A \cup B)$ and $P(A \cap B)$.

Hint. The power-of-two seeds are 1,2,4,8,.... Separate seed 1 from the remaining terms; all remaining power-of-two seeds are even.

★ 26. *Recovering atom probabilities.* A classifier produces one of four terminal states: correct-positive CP, correct-negative CN, false-positive FP, or false-negative FN. You are told $P(CP \cup CN)=0.82$, $P(CP \cup FP)=0.40$, $P(CP)=0.34$, and $P(FP \cup FN)=0.18$.

(a) Use disjointness of terminal states to recover all four atom probabilities.

(b) Verify the probability axioms for the recovered model.

(c) Find the probability of a negative prediction.

(d) Find the probability of an incorrect classification or a positive prediction.

Hint. Start with FP from $P(CP \cup FP)$, then FN from $P(FP \cup FN)$, and CN from $P(CP \cup CN)$. Check the total before answering the event questions.

★ 27. *No uniform distribution on all programs.* Imagine listing all syntactically valid finite programs as $p_1, p_2, \ldots$. Suppose someone proposes assigning the same probability q to every program.

(a) Show that $q > 0$ violates normalization.

(b) Show that $q=0$ also violates countable additivity and normalization.

(c) Conclude that no uniform probability distribution exists on a countably infinite set.

(d) Give one valid non-uniform probability assignment on this list.

Hint. For $q > 0$, a sufficiently large finite prefix already has total mass above 1. For $q=0$, sum the singleton probabilities using countable additivity.

★ 28. *Continuity from above for persistent failures.* Let F_n be the event that an automated repair system is still failing after n attempts. Then $F_1 \supseteq F_2 \supseteq \ldots$. Assume $P(F_1) < \infty$ and define $F=\cap_{n=1}^{\infty} F_n$.

(a) Apply continuity from below to the complements F_n^c to prove $P(F_n)$ tends to $P(F)$.

(b) Suppose $P(F_n)=1/(n+1)$. Find $P(F)$.

(c) Interpret F and the result in the repair-system context.

Hint. The complements form an increasing sequence. Use $P(F_n)=1-P(F_n^c)$ after taking the limit.